
What a Representation Should Keep, Ignore, or Refuse: Cue Reliance, Domain Shift and Open-Set Rejection

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Abstract

This project explores four ways in which the IID, closed-set assumption breaks, with one controlled protocol per task and seed 6304 throughout. In Task 1, on STL-10, frozen ResNet-50, ViT-B/16 and CLIP ViT-B/32 gave the same clean accuracy (96.4–97.6%) but relied on different evidence: CLIP is the most shape-driven on 200 AdaIN cue conflicts (shape bias 86%, coverage 70%), ResNet-50 the least (55%, 63%) and almost purely texture-driven on the bird–ship pair, while CLIP was the most hurt by patch shuffling (−14 pp). In Task 2, on PACS with Sketch as the unlabeled target, MMD alignment (DAN) lifts Sketch accuracy from 57.3% to 68.7%, both adversarial methods diverge under the prescribed settings, and a λ_{MMD} sweep shows that less domain separability came with worse target accuracy. In Task 3, without Sketch, pairwise source alignment (DAN-DG) made the sources way less separable but left Sketch unchanged (+0.5 pp), whereas SAM more than halves the local sharpness proxy and gains +11.0 pp, as much as target-aware DAN. In Task 4, on CIFAR-10 vs. CIFAR-100 unknowns, no score or model rejected more than a third of near unknowns at 95% known acceptance; RandAugment barely moves closed-set accuracy (+0.14 pp) or rejection, and selecting PROSER by closed-set accuracy stops it before its placeholders are learned. The common thread is that invariance and confidence only help when they are attached to the right cue. Code, configurations, split indices and all result files: <https://github.com/AsadAyub947/pa1-beyond-iid>.

1 Introduction and research questions

A 97% test accuracy tells us little about why a model is right. At deployment, an object can arrive with a new texture, position or color, the image distribution can shift, or the object can belong to an unknown class. So this report, throughout the four tasks, asks one question: what should a robust representation preserve, what should it become invariant to, and when should it refuse to answer? Task 1 tests which cues drive a CNN, a ViT and CLIP, and whether their features change the same way their predictions do. Task 2 tests whether aligning source and unlabeled Sketch features helps or removes class information. In Task 3, Sketch is removed entirely and ERM is compared with source-domain invariance (DAN-DG) and SAM. Task 4 tests whether a CIFAR-10 classifier can reject CIFAR-100 inputs and whether GCSC or PROSER help. The research questions of each task are answered at the end of Sections 3–6, and the cross-task questions in Section 7.

The main findings are that models with the same clean accuracy differ in shape bias and in sensitivity to global arrangement, and that CLIP’s shape preference is found in the frozen representation rather than the head. Secondly, on PACS, lower domain separability was not enough to get better target accuracy and was sometimes harmful, as marginal alignment imports the source label distribution. Thirdly, a flat solution like SAM transferred as well as target-aware alignment. Lastly, near unknowns receive near-certain known predictions, and under the selection rules neither augmentation nor placeholders change that.

2 Experimental setup

The data, models, optimizers and selection rule of each task are listed in Appendix Table 13; every run uses seed 6304.

Task 1: Inductive biases and feature representations. The three backbones are frozen and one linear head is trained on each (best validation accuracy 97.4, 97.9 and 98.6%). CLIP zero-shot uses the prompt “a photo of a {class}.” Each design choice has a hypothesis and a metric that were fixed before looking at any result. For STL-10, all backbones start from the same clean accuracy, so the later gaps show cue use and not head quality; the metrics are accuracy and macro-F1. For color, if color is a cue then a wrong color hurts as much as no color, so we use a fixed 90° hue rotation (it keeps brightness and shape and only changes the colors), and classes with typical coat colors should be hurt the most; the metrics are the change in accuracy and consistency. For cue conflicts (AdaIN [Huang and Belongie, 2017], style strength 1.0, style statistics from the central 70% crop, pairs monkey–ship, cat–monkey, airplane–truck, bird–ship, bird–dog, both directions), ResNet-50 should be more texture-based than ViT and CLIP the most shape-driven; the metrics are shape bias and coverage [Geirhos et al., 2019]. For the visualization (t-SNE, perplexity 30, PCA init, seed 6304, one joint fit per backbone and condition), color changes and shifts should keep points in their clean class clusters while cue conflicts and shuffling pull them out; the metric is how much the two conditions mix. The rejection rule was fixed before evaluation and only uses the AdaIN VGG encoder: reject degenerate outputs (pixel std < 0.02), outputs whose relu1_1–relu3_1 statistics are not closer to the style image than to the content image, and outputs whose relu4_1 layout is not closer to the content image. I also checked every contact sheet by eye (Appendix Figure 8). 200 images were accepted and 2 rejected (failed texture transfer), 20 per direction.

Tasks 2 and 3. Photo, Art painting and Cartoon are the labeled sources and Sketch is the target (PACS [Li et al., 2017]). All methods fine-tune the whole ResNet-18 with BatchNorm statistics frozen at their ImageNet values, with 8 images per source domain in every step (plus 24 Sketch images in Task 2). DAN [Long et al., 2015] adds an MMD penalty between source and Sketch features, DANN [Ganin et al., 2016] a domain discriminator behind a gradient reversal layer, and CDAN [Long et al., 2018] gives the discriminator the outer product of features and class probabilities. A step loss above 1000 counts as divergence. Both adversarial runs diverged, so they stay as the prescribed results, and two supplementary runs ([†]) only add a gradient penalty and gradient clipping. In Task 3, DAN-DG uses the same MMD code on every pair of sources and SAM [Foret et al., 2021] uses $\rho = 0.05$. Checkpoints are chosen by mean source-validation macro-F1 only, Sketch labels are read only by the final evaluation, and the Task 3 ERM model is the Task 2 Source-only checkpoint.

Task 4. The ten CIFAR-10 classes are known, and 800 near unknowns (bus, pickup truck, motorcycle, tractor, wolf, fox, leopard, camel) and 800 far unknowns (bottle, bowl, chair, clock, keyboard, mushroom, sunflower, wardrobe) come from the CIFAR-100 test set. On the frozen Vanilla ResNet-18 we compare maximum softmax (MSP), maximum logit (MLS), energy and a Mahalanobis distance. GCSC is the same recipe with RandAugment(2, 9). PROSER [Zhou et al., 2021] starts from Vanilla with five dummy classifiers and is fine-tuned for 50 epochs with classifier placeholders and data placeholders (manifold mixup after layer2). Every threshold τ is the 95th percentile of the score on CIFAR-10 validation data, and all checkpoints were fixed before any CIFAR-100 image was scored. Vanilla and GCSC were selected at epochs 98 and 94 of 100, PROSER at epoch 6 of 50.

3 Task 1: inductive biases and representations

3.1 Clean baseline and color

All four classifiers start from almost the same point, 96.4 to 97.6% accuracy with macro-F1 equal to accuracy (Table 1), so every intervention is read against each model’s own baseline. The CLIP head’s low confidence (0.654) comes from unit-norm embeddings and early stopping (Appendix Table 15), so confidences are not compared across heads. Color matters a little and about the same for all models: grayscale and the hue rotation both cost 2.8 to 4.6 pp, so a wrong color is not clearly worse than no color. The class-level part of the hypothesis did hold: under grayscale, dog is the worst or joint-worst class for all four classifiers (−10 to −14 pp), with flips to deer and horse. Hue rotation

Table 1: Clean baseline and color / patch-shuffle interventions on the 500-image subset (accuracy and Δ vs. the model’s own clean baseline in %, Cons = unchanged predictions in %, Conf = mean max softmax).

Model	Clean			Grayscale		Hue rotation 90°		Patch shuffle 4×4	
	Acc	F1	Conf	Acc (Δ)	Cons	Acc (Δ)	Cons	Acc (Δ)	Cons
ResNet-50	97.6	97.6	0.962	94.0 (-3.6)	95.2	94.8 (-2.8)	96.2	89.6 (-8.0)	90.6
ViT-B/16	97.6	97.6	0.973	94.8 (-2.8)	97.2	93.6 (-4.0)	95.2	90.8 (-6.8)	92.4
CLIP head	97.4	97.4	0.654	93.6 (-3.8)	94.0	92.8 (-4.6)	94.0	83.4 (-14.0)	84.0
CLIP zero-shot	96.4	96.4	0.936	91.8 (-4.6)	92.2	92.8 (-3.6)	93.4	81.4 (-15.0)	82.6

Table 2: Cue-conflict decisions (N = 200). Shape bias and coverage in %.

Model	N_{shape}	N_{texture}	N_{other}	Shape bias	Coverage
ResNet-50	69	57	74	54.8	63.0
ViT-B/16	104	27	69	79.4	65.5
CLIP head	121	19	60	86.4	70.0
CLIP zero-shot	124	21	55	85.5	72.5

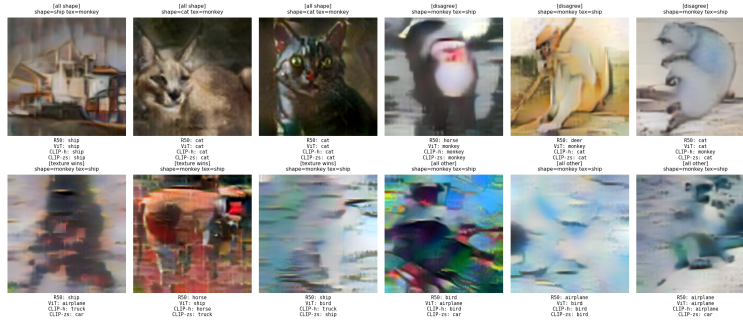


Figure 1: Cue-conflict examples with each model’s prediction. Top: all models follow the shape (ship, cat), and disagreements on monkey shapes with ship texture. Bottom: texture wins (ResNet says ship) and failures where no model gives either intended label.

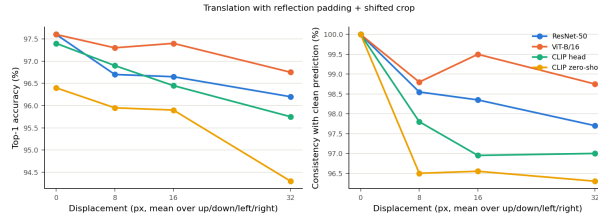


Figure 2: Accuracy and consistency vs. displacement (reflection padding and shifted crop, mean of four directions). Numbers per displacement are in Appendix Table 14.

also turns some ships into trucks for ResNet, so the blue of water seems to be part of what ResNet calls a ship.

3.2 Shape versus texture

The ordering matches the hypothesis: CLIP (86.4% head, 85.5% zero-shot) is ahead of ViT (79.4%) and ResNet-50 (54.8%) (Table 2). But coverage is only 63 to 72.5%, and the other decisions are not random (ResNet sends them to deer and bird, ViT to airplane). The pairs tell more than the total. ResNet-50 is shape-driven on airplane–truck, bird–dog and cat–monkey (77 to 80%) but follows texture on bird–ship (10%, coverage 78%): 14 of 20 ship shapes with bird texture become bird, and 14 of 20 bird shapes with ship texture become ship. Since it happens in both directions, it is a real texture preference and not a class prior. Monkey–ship has only 20 to 48% coverage for every model because the ship style damaged both cues (Figure 1), so I do not read anything into that pair.

Table 3: Cosine stability I_T between clean and transformed features, with each backbone’s own clean-image baselines (random pair and same-class pair). Cue conflicts are compared with their content image.

Backbone	Cosine stability I_T							Clean-image baselines	
	Gray	Hue	Shuffle	Cue conf.	T 8px	T 16px	T 32px	Random pair	Same class
ResNet-50	0.798	0.749	0.692	0.293	0.967	0.951	0.931	0.222	0.454
ViT-B/16	0.732	0.716	0.628	0.274	0.945	0.976	0.944	0.069	0.291
CLIP ViT-B/32	0.888	0.884	0.745	0.701	0.978	0.963	0.967	0.592	0.737



Figure 3: t-SNE, one joint fit per backbone and condition. Top: patch shuffle, bottom: cue conflict; left to right ResNet-50, ViT-B/16, CLIP. Color = class, filled = clean, hollow = transformed.

3.3 Spatial structure: translation and patch shuffling

All models are close to translation-invariant at these shifts (Figure 2): the largest drop at 32 px is 2.1 pp and consistency stays above 96%. ResNet-50 decays smoothly, but ViT-B/16 does not: its consistency is 98.8, 99.5 and 98.8% at 8, 16 and 32 px, and its feature stability goes up from 0.945 to 0.976 between 8 and 16 px (Table 3). A 16 px shift moves the image by exactly one token, so each patch embedding stays almost the same, while an 8 px shift cuts every token in half. CLIP B/32 shows the same effect, weaker, at 32 px. Patch shuffling is much harsher (Table 1). ResNet (−8.0 pp) and ViT (−6.8 pp) still get about 90% of shuffled images right with confidence around 0.91, so a lot of STL-10 can be classified from local texture and parts. CLIP loses 14 to 15 pp, mostly inside superclasses (shuffled dogs become cats, ships become airplanes).

3.4 Representation analysis

Raw cosines cannot be compared across backbones, because two random clean images already have cosine 0.59 under CLIP but 0.07 under ViT (Table 3). Measured against each backbone’s random-pair baseline, the backbones react almost the same to color and translation, while shuffling moves CLIP much further: it keeps $(0.745 - 0.592)/(1 - 0.592) = 0.38$ of the way from random to identical, against 0.60 for ResNet and ViT. The t-SNE plots agree (Figure 3; grayscale and translation in Appendix Figure 9): color changes and shifts stay on the clean clusters, cue conflicts form a separate, poorly separated region, and shuffled CLIP features leave the clean clusters and regroup by superclass.

3.5 Answers to the Task 1 research questions

RQ1: shape, texture and color. Color is a secondary cue for all models: it costs a few points, mostly in classes with typical coat colors. Texture is different: of the covered conflicts ResNet-50 resolves 45% by texture, ViT 21% and CLIP 14%. The ranking CLIP, ViT, ResNet holds on four of the five pairs, which I trust more than one total number.

RQ2: spatial structure. Small shifts are handled by all three models; what differs is how invariance breaks. ResNet decays smoothly with distance, while both ViTs change with their token grid, so they are sensitive to grid alignment rather than object position. Global layout matters less than I expected

for ResNet and ViT, which is also why ResNet can rely on texture at no cost on clean data. CLIP depends more on global arrangement, which fits a model trained to match whole-image captions.

RQ3: predictions versus representations. The two levels agree in ranking but not in detail. Under grayscale, ViT features move more than ResNet features (0.71 vs. 0.74 after rescaling) while ViT predictions move less (97.2 vs. 95.2% consistency), so the change lies mostly outside what the ViT head reads. Cue conflicts are the extreme case: ViT’s stylized features are less similar to their own content image (0.274) than two clean images of the same class (0.291), yet the head still returns the content label for 52% of them, so a small class subspace survives and a linear head only needs that. CLIP zero-shot and the CLIP head agree on only 71.0% of cue conflicts (Appendix Table 16) but have almost the same shape bias (85.5 vs. 86.4%), so CLIP’s shape preference comes from the representation and not the head.

RQ4: architecture versus training. Two differences can be put on architecture with some confidence: the token-aligned translation effect in both ViTs and ResNet’s smooth local decay. The shape-bias ranking cannot. ViT-B/16 and CLIP share the transformer design but differ by 7 pp in shape bias and 7–8 pp in shuffle robustness, and they also differ in data, supervision, patch size and model size. Since the CLIP head and zero-shot share the same shape bias, pretraining is the most likely source.

4 Task 2: unsupervised domain adaptation

DAN trained as intended: its source F1 stays at the Source-only level while the MMD falls from 0.067 to about 0.011 (Figure 4). DANN never learned the class task: its feature norms were far above the others from epoch 0 (336 vs. 44 to 62), its source F1 peaked at 0.26, and at step 929 the loss passed 1000 (Appendix Table 17). CDAN diverged at step 200 and put every Sketch image in one class. So a discriminator near 0.5 is not proof of alignment here: with frozen BatchNorm statistics the reversed gradient can beat the discriminator by just scaling the features up, and the linear probe still separates the domains at 97.5 and 99.7% (Table 4). The stabilized DANN still diverged, and the stabilized CDAN trained but ended 14.1 pp below Source-only (Appendix Figure 12).

At class level (Table 5, Figure 5), Source-only turns most sketched dogs and horses into elephants and giraffes. DAN fixes a large part of this but sends 300 dogs to person and predicts person and house far more often than they occur. The class shares explain why (Appendix Table 18): person and house are about 22% and 14% of the source images but only 4% and 2% of Sketch.

For the controlled study I expected more alignment pressure to lower separability, cost a little source accuracy, and help Sketch up to some point. The first two held (separability 97.0 to 82.3%, mean source accuracy 94.6 to 89.8%, Table 6), but Sketch accuracy fell the whole way from 76.3 to 61.0%, so the weakest alignment was the best.

4.1 Answers to the Task 2 research questions

RQ1: domain gap. Source-only reaches 94.9% mean source-validation accuracy and 57.3% on Sketch, a gap of 37.6 pp. The failures sit in two classes, dog (13.0%) and horse (21.2%). Elephant (95.5%) and giraffe (86.7%) look good only because they absorb everything else: the model predicts elephant for 1505 and giraffe for 1377 of the 3929 sketches. Sketches keep contours and drop color and texture, so animals with a distinctive outline (trunk, long neck) survive, while dogs and horses, which photos separate mostly by fur color and texture, fall to the nearest large-animal outline.

RQ2: separability versus recognition. The only successful method, DAN, also has the lowest separability (87.0%), which first looks like support for alignment, but the class-level results say otherwise. DAN’s +11.4 pp comes from dogs (+21.5), horses (+46.2) and the rare classes, but it costs elephant (−17.6) and giraffe (−15.8), and person and house are now predicted 685 and 369 times with precision of only 18.7% and 21.7%. This is the label-shift problem described by Zhao et al. [2019]: matching marginal feature distributions pushes sketched animals into the source regions of person and house, so there is class-level negative transfer hidden inside a positive total.

Table 4: Source-validation (P/A/C) and Sketch results in %; Δ vs. Source-only; Sep. = source-vs-Sketch domain separability (chance 50); Div. = run diverged. $\dagger$ supplementary stabilized runs, not part of the prescribed comparison.

Method	Source val acc			Mean source		Sketch (target)		Δ Acc	Sep.	Div.
	P	A	C	Acc	F1	Acc	F1			
Source-only	96.4	94.2	94.2	94.9	94.9	57.3	58.4	+0.0	99.7	no
DAN	96.4	92.9	93.8	94.4	94.3	68.7	63.3	+11.4	87.0	no
DANN	45.1	22.2	32.4	33.2	25.7	30.9	20.8	-26.4	97.5	yes
CDAN	43.5	26.3	22.6	30.8	21.1	2.0	0.6	-55.3	99.7	yes
DANN-stab. $\dagger$	88.9	73.7	72.3	78.3	77.2	15.5	3.8	-41.8	99.9	yes
CDAN-stab. $\dagger$	96.7	90.0	91.3	92.7	92.8	43.2	48.6	-14.1	98.6	no

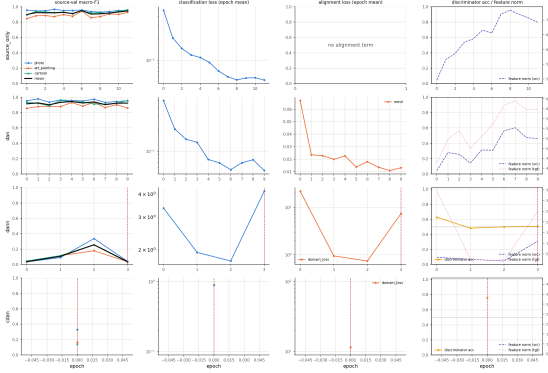


Figure 4: Per-epoch source-validation F1, classification loss, alignment loss and discriminator accuracy / feature norm for the four prescribed runs; red dashed line = divergence detected.

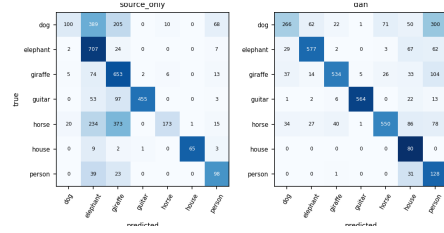


Figure 5: Sketch confusion matrices of Source-only and DAN (counts; color = row-normalized). All six runs: Appendix Figure 11.

Table 5: Per-class Sketch accuracy of Source-only, change for each method (pp), and number of Sketch images predicted as each class.

Class	n	Source-only		DAN		DANN		CDAN	
		Acc	#pred	Δ Acc	#pred	Δ Acc	#pred	Δ Acc	#pred
dog	772	13.0	127	+21.5	367	-12.0	16	-13.0	0
elephant	740	95.5	1505	-17.6	682	-33.8	1200	-95.5	0
giraffe	753	86.7	1377	-15.8	605	-64.1	434	-86.7	0
guitar	608	74.8	458	+17.9	571	-74.5	2	-74.8	0
horse	816	21.2	189	+46.2	650	+40.2	1359	-21.2	0
house	80	81.2	66	+18.8	369	-77.5	3	+18.8	3929
person	160	61.3	207	+18.8	685	-14.4	915	-61.3	0

Table 6: Controlled study, $\lambda_{\text{MMD}} \in \{0.1, 1, 10\}$ ($\lambda = 1$ is the main DAN run). Sketch columns were computed after the grid was fixed.

λ_{MMD}	P	A	C	Mean acc	Mean F1	Sketch acc	Sketch F1	Sep.
0.1	97.0	91.5	95.3	94.6	94.6	76.3	71.4	97.0
1	96.4	92.9	93.8	94.4	94.3	68.7	63.3	87.0
10	93.4	86.1	89.8	89.8	89.3	61.0	56.5	82.3

RQ3: conditional versus marginal alignment. CDAN did not keep the semantic structure better than DAN; it did worse. The prescribed run diverged, and the stabilized run lost 14.1 pp, with elephant (-81.9 pp) and giraffe (-40.9 pp) collapsing into person, which it predicts 1735 times for 160 real persons (Appendix Figure 14). Conditioning on class probabilities only helps if they are roughly right on the target, but the starting classifier calls 73% of Sketch elephant or giraffe, so CDAN aligned Sketch modes with the wrong source modes and amplified the source prior.

RQ4: alignment strength. Stronger alignment made the domains harder to separate and lowered source accuracy as expected, but it lowered Sketch accuracy too: alignment helps while it gently moves the features and hurts once it starts moving the classes. Mean source-validation F1 also ranks $\lambda = 0.1$ first (94.6 vs. 94.3 vs. 89.3), so the best setting could have been chosen without target labels, although the main comparison uses $\lambda = 1$ as required.

5 Task 3: domain generalization

DAN-DG makes the sources much less separable (86.0 to 68.7%) and its pairwise MMD falls smoothly (Table 7, Figure 6), but Sketch moves by only +0.5 pp. SAM more than halves the sharpness proxy (0.135 vs. 0.318 for ERM) and gains +11.0 pp on Sketch.

At class level (Table 8), SAM mostly repairs horse (+41.8) and dog (+32.1) but loses giraffe (−25.2) and person (−25.6). DAN-DG gains on dog, horse and person but loses house (−28.7) and elephant (−23.1), and like DAN it predicts person far too often (921 times). Both alignment methods help dogs and horses, but only Task 2 DAN repairs house and guitar.

For the study I expected larger λ_{DG} to lower separability, cost some source accuracy, and help Sketch only at moderate values. The first two held and the third did not (Table 9): Sketch accuracy falls as λ_{DG} grows, and $\lambda = 0.1$ gives the best Task 3 result (72.3%) with separability (85.7%) almost equal to ERM’s. At $\lambda = 10$ the model never learns (training loss stays near $\ln 7$), and separability drops to 54% only because the features collapsed.

5.1 Answers to the Task 3 research questions

RQ1: do source metrics predict Sketch? Only weakly. Mean-source accuracy puts SAM first (95.7%), in line with Sketch, but worst-source accuracy puts ERM first (94.2) although ERM is last on Sketch, and in the study DAN-DG with $\lambda = 0.1$ has lower mean-source accuracy than ERM (94.0 vs. 94.9) but is 15 pp better on Sketch. Among the models that trained, source metrics span 2.2 pp while Sketch spans 15 pp, so source validation only catches a collapse, and it cannot see that ERM gets only 13% of sketched dogs and 21% of horses right.

RQ2: source invariance. DAN-DG makes the sources less distinguishable, but that invariance did not transfer and seems to have removed class information: Art painting, the most distinct source, lost 4.4 pp of validation accuracy, and on Sketch house and elephant dropped while person became a sink again. At $\lambda = 10$ both separability and the classifier are near chance, so the most invariant model is also the most useless one.

RQ3: sharpness. SAM lowers the proxy to less than half of ERM (0.318) and DAN-DG (0.347) and is the only method that clearly improves Sketch. The rankings by sharpness and by Sketch agree only at the top, and the study breaks the link: $\lambda = 0.1$ and $\lambda = 1$ have the same sharpness (0.349 vs. 0.347) but differ by 15 pp on Sketch. Also, SAM minimizes almost the quantity the proxy measures, and it kept improving until epoch 15 while ERM stopped at epoch 6.

RQ4: value of unlabeled Sketch data. With the same ERM baseline and MMD code, target-aware DAN gains +11.4 pp and target-free DAN-DG +0.5 pp (Appendix Table 19), and only DAN repairs the rare classes (house +18.8 vs. −28.7). So seeing unlabeled sketches looks worth about 11 points, but this is not a clean causal result: the objectives and batch sizes differ, the protocol is transductive, and SAM reaches the same +11 pp without any target data.

6 Task 4: open-set recognition

On the Vanilla model (94.82% closed-set accuracy), no score rejects more than a third of near unknowns while keeping about 95% of known test images (Table 10). MSP has the best near AUROC (82.1) and Mahalanobis the best far AUROC (91.9), but energy rejects the most at the operating point (33.0% near, 54.4% far), and near unknowns overlap the known scores for every plotted score (Figure 7).

Table 7: Source-validation and Sketch results (%); Δ vs. ERM; Sep. = 3-way source-domain separability (chance 33.3); Δ_{sharp} = common sharpness proxy (CE increase after one ascent step of radius 0.05). Worst domains: ERM: art painting; DAN-DG: art painting; SAM: art painting.

Method	Source val acc			Mean source		Worst source		Sketch		ΔAcc	Sep.	Δ_{sharp}
	P	A	C	Acc	F1	Acc	F1	Acc	F1			
ERM	96.4	94.2	94.2	94.9	94.9	94.2	94.2	57.3	58.4	+0.0	86.0	0.318
DAN-DG	96.7	89.8	94.0	93.5	93.3	89.8	89.5	57.8	57.7	+0.5	68.7	0.347
SAM	98.2	92.7	96.2	95.7	95.8	92.7	92.8	68.3	67.5	+11.0	87.7	0.135

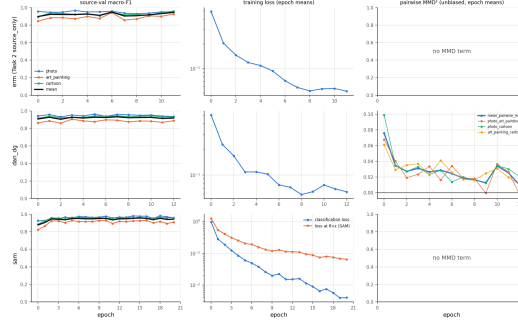


Figure 6: Training curves: source-validation F1, classification loss (and the loss at $\theta + \epsilon$ for SAM), and the pairwise MMD penalty of DAN-DG.

Table 8: Per-class Sketch accuracy of ERM, changes for DAN-DG and SAM, and the change of Task 2 DAN for comparison (pp).

Class	n	ERM		DAN-DG		SAM		Task 2 DAN
		Acc	#pred	ΔAcc	#pred	ΔAcc	#pred	ΔAcc
dog	772	13.0	127	+12.6	327	+32.1	520	+21.5
elephant	740	95.5	1505	-23.1	743	-4.0	1322	-17.6
giraffe	753	86.7	1377	-6.2	1043	-25.2	594	-15.8
guitar	608	74.8	458	-2.3	446	+15.8	571	+17.9
horse	816	21.2	189	+17.0	406	+41.8	715	+46.2
house	80	81.2	66	-28.7	43	+11.2	106	+18.8
person	160	61.3	207	+23.1	921	-25.6	101	+18.8

Table 9: Controlled study, $\lambda_{\text{DG}} \in \{0.1, 1, 10\}$ ($\lambda = 1$ is the main DAN-DG run). Sketch values are for analysis only.

λ_{DG}	Mean acc	Mean F1	Worst acc (domain)	Sketch acc	Sketch F1	Sep.	Δ_{sharp}
0.1	94.0	94.1	91.0 (art painting)	72.3	74.5	85.7	0.349
1	93.5	93.3	89.8 (art painting)	57.8	57.7	68.7	0.347
10	25.6	15.1	18.1 (cartoon)	16.9	10.1	54.0	2.208

For the failure analysis I looked at the accepted images myself (Table 12, Appendix Figure 17). Near cases such as camel to deer, pickup truck to automobile, fox to dog and bus to truck are plausible, and tractor to ship (sandy shore) or leopard to frog (spotted skin) come from background or texture. Far labels are all surprising: wardrobe to truck (box shape) and chair to airplane (thin struts on a white sky) can be explained, but mushroom to bird, keyboard to bird and bottle or bowl to cat cannot.

Positive augmentation versus placeholders. GCSC and PROSER try to help in opposite ways. RandAugment makes the known classes survive distortions, which raised closed-set accuracy by only 0.14 pp and rejection by 0.6 pp (near) and 2.5 pp (far). PROSER instead teaches the model where unknowns might be. Its classifier placeholder is learned almost at once (from epoch 2 the dummy is the runner-up for over 99.8% of training images), but the data placeholders only take hold after epoch 8, when the share of mixed features sent to the dummy rises from 18% to about 99%, and validation accuracy then falls to 85–95% and never returns to its epoch-6 value. The trade-off is real: selection by validation accuracy picks epoch 6, where the placeholders are barely trained, so PROSER lost 0.63 pp of accuracy without gaining any rejection.

Table 10: Post-hoc scores on the frozen Vanilla model (%); τ = 95th percentile of u on CIFAR-10 validation; Known kept = known test images accepted at τ ; FPR@95TPR = fraction of unknowns accepted.

Score	AUROC				Rejection @ τ			FPR@95TPR	
	Near	Far	All	Known kept	Near	Far	All	Near	Far
MSP	82.1	89.8	85.9	94.6	27.9	43.6	35.8	72.1	56.4
MLS	80.6	89.4	85.0	95.1	31.9	53.0	42.4	68.1	47.0
Energy	80.7	89.4	85.1	95.3	33.0	54.4	43.7	67.0	45.6
Mahalanobis	80.5	91.9	86.2	94.8	28.0	52.2	40.1	72.0	47.8

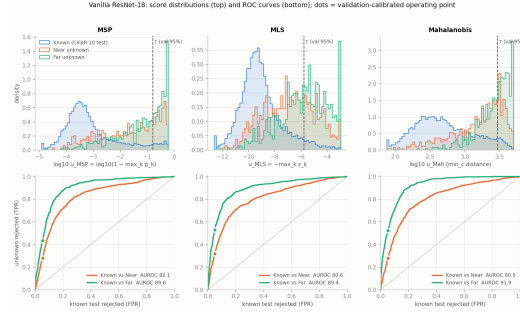


Figure 7: Score distributions (top) and ROC curves (bottom) for MSP, MLS and Mahalanobis on the Vanilla model; dots mark the validation-calibrated operating point.

Table 11: Trained-model comparison with MLS as the common score and the PROSER placeholder scores (CSA from the ten known logits).

Model	Score	CSA	AUROC				Rejection @ τ			FPR@95TPR	
			Near	Far	All	Known kept	Near	Far	All	Near	Far
Vanilla	MLS	94.82	80.6	89.4	85.0	95.1	31.9	53.0	42.4	68.1	47.0
GCSC	MLS	94.96	81.4	89.8	85.6	95.3	32.5	55.5	44.0	67.5	44.5
PROSER	MLS	94.19	79.8	89.6	84.7	95.4	27.6	46.8	37.2	72.4	53.2
PROSER	PROSER dummy	94.19	79.3	88.5	83.9	95.5	29.0	45.4	37.2	71.0	54.6
PROSER	PROSER margin	94.19	79.3	88.4	83.9	95.4	29.1	46.0	37.6	70.9	54.0

Table 12: Wrongly accepted unknowns at the Vanilla MLS threshold (the four most confident per group). The images are in Appendix Figure 17.

Group	Unknown class	Predicted	u_{MLS}	τ	Max softmax	A-priori label
near	camel	deer	-11.89	-5.78	1.0000	plausible
near	pickup truck	automobile	-11.67	-5.78	1.0000	plausible
near	fox	dog	-11.67	-5.78	0.9998	plausible
near	bus	truck	-11.35	-5.78	0.9998	plausible
far	mushroom	bird	-12.37	-5.78	1.0000	surprising
far	keyboard	bird	-12.24	-5.78	1.0000	surprising
far	bottle	cat	-11.53	-5.78	0.9999	surprising
far	bowl	cat	-10.95	-5.78	0.9999	surprising

6.1 Answers to the Task 4 research questions

RQ1: semantic similarity. Semantic closeness decides most of it. At the Vanilla MLS threshold 68% of near but 47% of far unknowns are accepted, and near AUROC is 7.7 to 11.4 pp below far AUROC for every score. The most accepted near classes are pickup truck (89%, mostly called automobile), bus (86%, truck), wolf (78%, cat or dog), and fox and leopard (70%, mainly cat) (Appendix Table 20), which are failures a person would forgive. Motorcycle, with no close CIFAR-10 class, is the best-rejected near class (36% accepted). Far unknowns are accepted less often but get stranger labels, and some failures show the cue that was used, like a leopard taken for a frog because of its spots, the same texture shortcut as in Task 1.

RQ2: what each score captures. MSP, MLS and energy all read the logits, and MLS and energy are almost the same score (Spearman 1.00, Appendix Table 21) because one logit dominates the

sum for a confident model. MSP removes the logit magnitude and saturates, so it has the best near AUROC but the worst rejection at the operating point (27.9 / 43.6%): many unknowns share the $1 - p \approx 0$ region with the knowns, and keeping the magnitude rejects 4 to 11 pp more. Mahalanobis measures something else (rank correlation with MLS 0.43): it accepts only 38% of wardrobes, which give a strong truck logit far from the truck features, but more motorcycles and tractors, which really sit near the vehicle clusters. So logit magnitude is the better signal when an unknown looks like a known class, and feature distance when it only triggers one class feature.

RQ3: GCSC. GCSC did not improve in a meaningful way (Table 11): closed-set accuracy moved by +0.14 pp and the AUROC gains are under 1 pp. This agrees with Vaze et al. [2022]: open-set performance follows closed-set accuracy, which barely moved because a 100-epoch ResNet-18 is already near its ceiling (training accuracy 99.99%). The small gain is on the far side, where augmentation weakens accidental single-cue activations, but no augmentation of dog images teaches that a wolf is not a dog. GCSC even made Mahalanobis worse on near unknowns (AUROC 77.2 vs. 80.5), probably because more varied training images widen the known clusters.

RQ4: PROSER. PROSER did not improve rejection beyond Vanilla or GCSC: with MLS it rejects 4.3 pp fewer near and 6.3 pp fewer far unknowns, and its placeholder score is slightly worse (AUROC 79.3 / 88.5). Even trained placeholders would likely help little, since a mixed dog/cat feature lies between two known classes, while a wolf lies beyond dog and a sunflower between no pair at all.

7 Cross-task discussion

Visual cues and distribution shift. The cue sensitivities from Task 1 line up with the PACS failures, as a pattern and not a proven cause. Under grayscale dog was the most damaged STL-10 class and ResNet-50 followed texture on bird-ship, and Sketch removes exactly these cues: ERM loses dogs and horses on Sketch to elephants and giraffes, whose outlines are distinctive.

Invariance versus discriminability. Making domains less distinguishable did not by itself improve recognition: inside each sweep Sketch accuracy falls as separability falls, DANN and CDAN fooled their discriminators while staying 97–99.7% separable, and the most invariant model (DAN-DG, $\lambda = 10$) had collapsed, while the mild $\lambda = 0.1$ settings helped without changing separability much.

The importance of target-domain access. Target-aware DAN beat target-free DAN-DG by 11 points with the same MMD code, but target access was not necessary (SAM +11.0 pp, DAN-DG with $\lambda = 0.1$ +15.0 pp), and it added risk: the adversarial methods collapsed onto one class and marginal alignment imported the source label prior.

Recognition versus rejection. GCSC improved closed-set accuracy and rejection slightly and PROSER lowered both, but neither made the model safer on near unknowns: every model and score accepts 69 to 94% of wolves, foxes, buses and pickup trucks. A classifier should abstain on close relatives of known classes and on inputs that trigger one known cue with nothing behind it, like a keyboard read as a bird.

Overall synthesis. A robust representation should keep shape and part structure, the only cues that survived every shift here, and be invariant to color, texture, rendering style and small translations but not to global arrangement. Invariance should be encouraged gently (weak MMD, SAM) rather than forced, and the model should refuse when its logit magnitude is low or its features are far from every class.

8 Conclusion

In all four tasks the overall number hid what the model was doing: the same clean accuracy hid a 30-point spread in shape bias, a discriminator at chance hid exploding features, a positive PACS total hid class-level negative transfer, and 95% closed-set accuracy hid confident acceptance of wolves as dogs. The main limitations are one seed per setting, upsampled STL-10 images, two diverged adversarial methods and proxy diagnostics.

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A Implementation details

Cue conflicts use the pretrained AdaIN decoder and VGG encoder from the public pytorch-AdaIN release; CLIP is loaded as ViT-B-32-quickgelu, the activation the OpenAI weights were trained with. MMD (Tasks 2 and 3) uses three RBF kernels with bandwidths 0.5, 1 and 2 times the median squared distance and the unbiased estimator, because the biased one has a floor of about 0.16 for 24 vs. 24 samples. SAM is wrapped around AdamW. The Mahalanobis score uses one shared diagonal covariance. PROSER uses half of each batch for the classifier placeholders ($\beta = 1$) and half for data placeholders made by manifold mixup after layer2 between different classes ($\gamma = 0.1$); its detection score is the softmax mass of the strongest dummy logit, as in the authors’ code, and the “margin” row in Table 11 uses the paper’s bias calibration (dummy logit minus the largest known logit), with the threshold set on CIFAR-10 validation data. The two supplementary Task 2 runs add only a DomainBed-style gradient penalty (weight 1) on the discriminator input and gradient clipping at 5. Tasks 2 and 3 run with deterministic cuDNN and seeded data loaders, and the Task 3 ERM checkpoint reproduces the Task 2 Source-only numbers exactly. External code (AdaIN weights, OpenCLIP, the DomainBed gradient penalty, the PROSER reference code) is credited in the repository README.

B Additional results

Table 13: Shared protocol per task. Every run uses seed 6304.

	Task 1	Tasks 2 & 3	Task 4
Data	STL-10: 80/20 split of train (4000/1000), 500-image test subset (50/class)	PACS, sources P/A/C (80/20 per domain), target Sketch (3929)	CIFAR-10 90/10 split (45k/5k), full test; CIFAR-100 test near/far (800 each)
Model	frozen ResNet-50 (V2), ViT-B/16 (V1), CLIP ViT-B/32 (openai) + linear head	ResNet-18 (ImageNet V1), full fine-tuning, BN statistics frozen	CIFAR ResNet-18 (3×3 stem, no max-pool), from scratch
Optimiser	AdamW, lr 0.001, wd 0.0001	AdamW, lr 0.0001, wd 0.0001	SGD, lr 0.1, momentum 0.9, wd 0.0005, cosine; PROSER lr 0.001
Budget	≤50 epochs, patience 5, batch 32	≤30 epochs, patience 5, 8/domain + 24 target	100 epochs (PROSER 50), batch 128
Selection	val accuracy	mean source-val macro-F1	CIFAR-10 val accuracy
Seed	6304	6304	6304

Table 14: Translation accuracy / consistency per displacement (%), mean of four directions.

	0 px		8 px		16 px		32 px	
Model	Acc	Cons	Acc	Cons	Acc	Cons	Acc	Cons
ResNet-50	97.6	100.0	96.7	98.5	96.7	98.4	96.2	97.7
ViT-B/16	97.6	100.0	97.3	98.8	97.4	99.5	96.8	98.8
CLIP head	97.4	100.0	96.9	97.8	96.5	97.0	95.8	97.0
CLIP zero-shot	96.4	100.0	96.0	96.5	95.9	96.5	94.3	96.3

Table 15: Linear-head training (Task 1).

Backbone	Best val acc	Best epoch	Epochs run
ResNet-50	97.40	7	13
ViT-B/16	97.90	5	11
CLIP head	98.60	6	12

Table 16: CLIP zero-shot vs. trained head agreement (%).

Condition	clean	grayscale	hue rotation	patch shuffle	cue conflict
Zero-shot vs. head agreement	96.2	94.6	94.4	85.8	71.0



Figure 8: Example contact sheet (content, style, conflict) used for the visual check before evaluation.

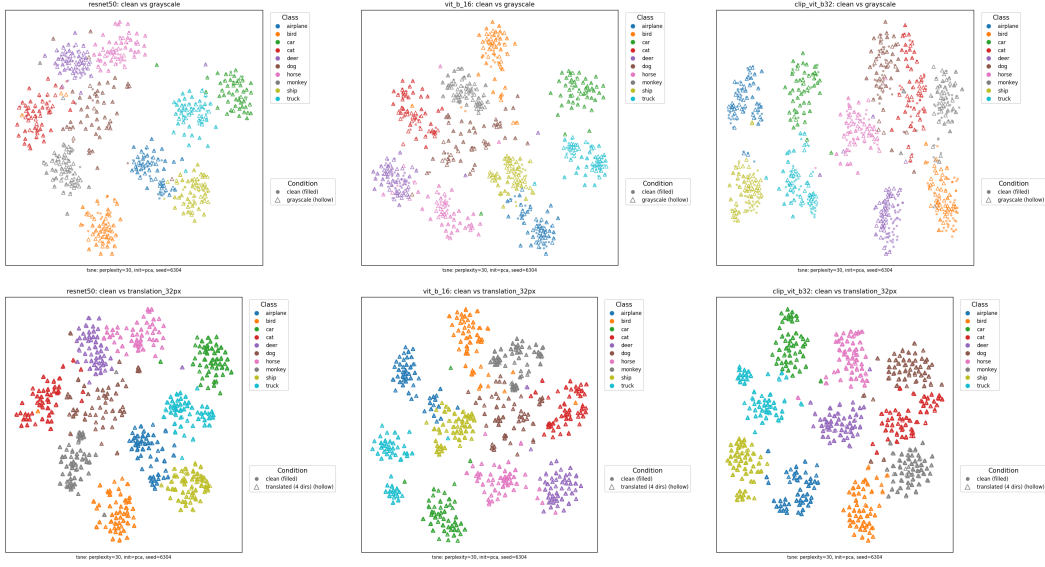


Figure 9: t-SNE (perplexity 30, PCA init, seed 6304) for grayscale (top) and 32 px translation, all four directions (bottom); left to right ResNet-50, ViT-B/16, CLIP. The transformed points stay on the clean clusters. Coordinates are not comparable across panels.

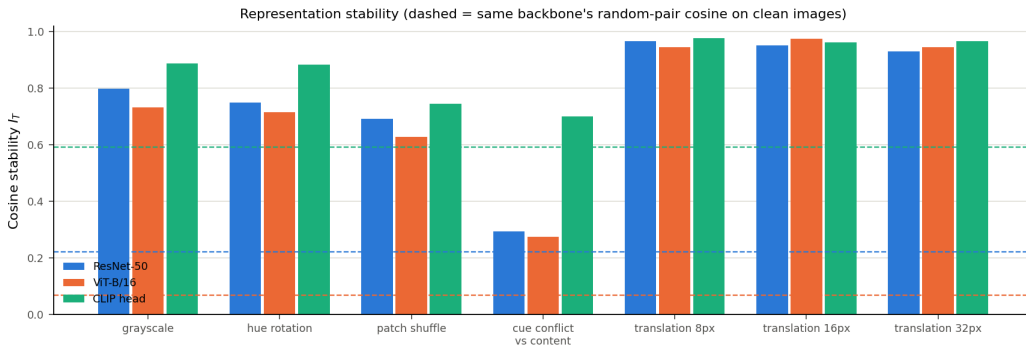


Figure 10: I_T bars with each backbone's random-pair baseline (dashed).

Table 17: Step at which each adversarial run passed the divergence threshold (loss > 1000), with feature norms and discriminator accuracy at that step. Epoch-0 mean source feature norms for reference: Source-only 49.6; DAN 44.4; DANN 336.3; CDAN 62.4.

Run	Epoch	Step	Step loss	α	$\ f_s\ $	$\ f_t\ $	Disc. acc
DANN	3	929	1546	0.58	1.85e+05	2.8e+05	0.50
CDAN	0	200	1382	0.14	2.26e+03	5.73e+04	0.31
DANN-stab. [†]	1	414	1027	0.29	7e+04	2.5e+05	0.33
CDAN-stab. [†]				did not diverge			

Table 18: Class distribution (%) of each PACS domain.

Domain	dog	elephant	giraffe	guitar	horse	house	person
Photo	11.3	12.1	10.9	11.1	11.9	16.8	25.9
Art	18.5	12.5	13.9	9.0	9.8	14.4	21.9
Cartoon	16.6	19.5	14.8	5.8	13.8	12.3	17.3
Sketch	19.6	18.8	19.2	15.5	20.8	2.0	4.1

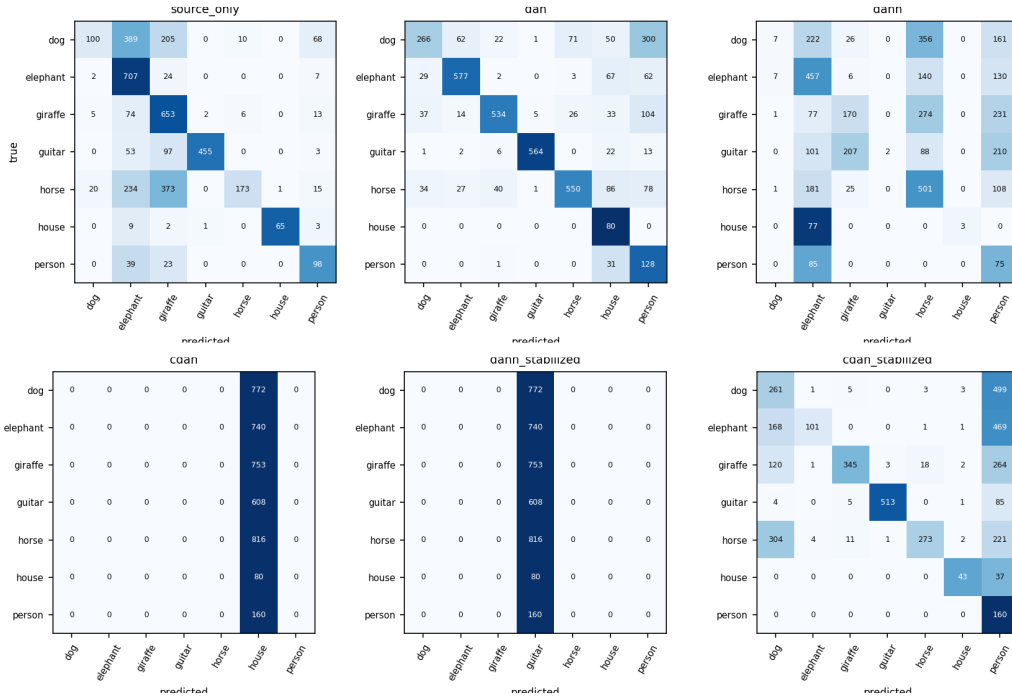


Figure 11: Sketch confusion matrices of all six Task 2 runs (counts; color = row-normalized). CDAN puts every sketch in house and the stabilized DANN puts every sketch in guitar.

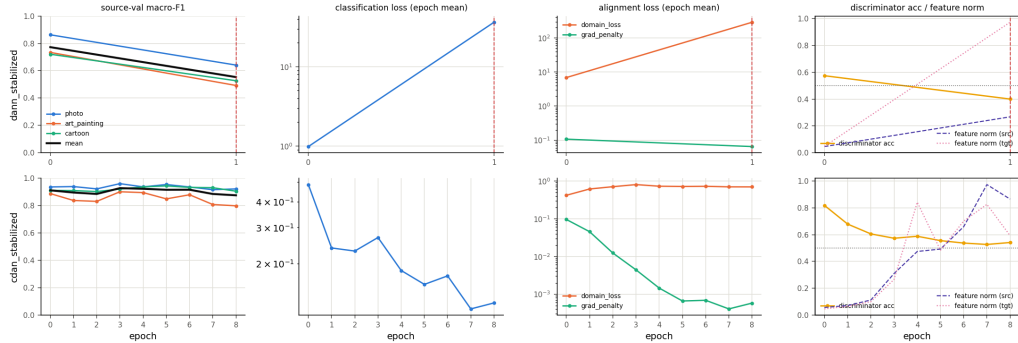


Figure 12: Training curves of the supplementary stabilized runs (DANN[†] top, CDAN[†] bottom).

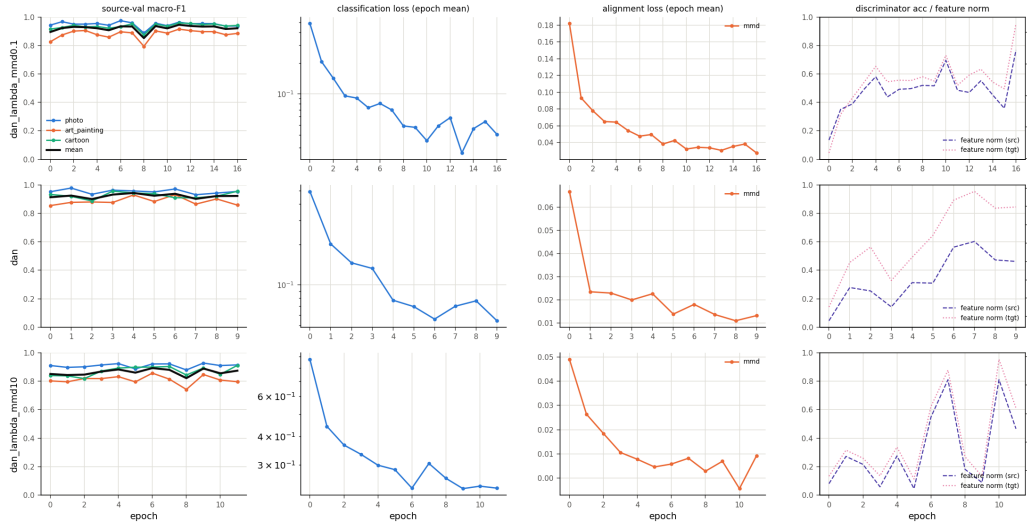


Figure 13: Training curves of the λ_{MMD} study ($\lambda = 0.1, 1, 10$ from top to bottom).

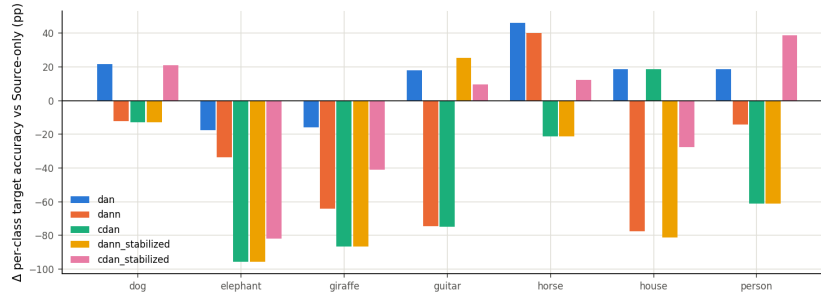


Figure 14: Per-class Sketch accuracy change vs. Source-only for every Task 2 run, including the stabilized ones.

Table 19: Same ERM baseline, same MMD code, different information (Task 2 DAN sees unlabeled Sketch; DAN-DG does not). The two separability scores measure different things.

Model	Sketch acc	Δ vs. ERM	Separability (what it separates)
ERM (Task 2 Source-only)	57.3	–	99.7 (src vs. Sketch); 86.0 (among src)
DAN (Task 2, sees unlabeled Sketch)	68.7	+11.4	87.0 (src vs. Sketch)
DAN-DG (Task 3, no Sketch)	57.8	+0.5	68.7 (among src)

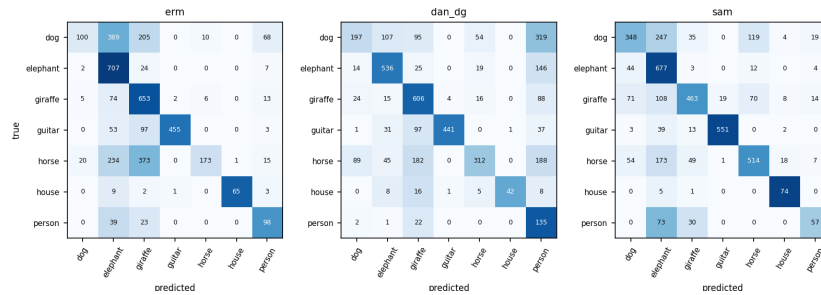


Figure 15: Sketch confusion matrices for ERM, DAN-DG and SAM.

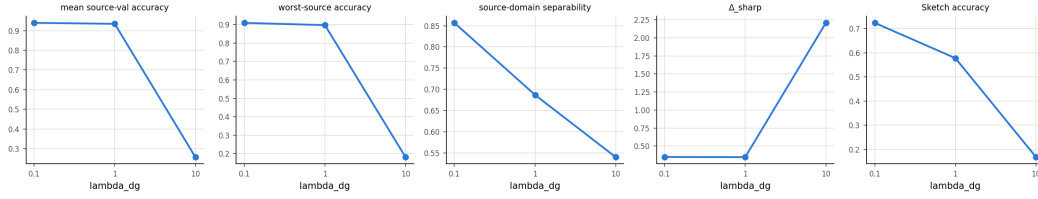


Figure 16: λ_{DG} study: source accuracy, worst-source accuracy, separability, sharpness and Sketch accuracy.



Figure 17: Accepted near (top) and far (bottom) unknowns at the Vanilla MLS threshold ($\tau = -5.78$), with predicted class and score.

Table 20: Acceptance rate (%) of each unknown class at the Vanilla MLS threshold and the CIFAR-10 class that absorbs most of them (share of accepted, %).

Near unknowns			Far unknowns		
Class	Accepted	Absorbed by (share)	Class	Accepted	Absorbed by (share)
pickup truck	89	automobile (72)	wardrobe	68	truck (51)
bus	86	truck (79)	sunflower	52	bird (42)
wolf	78	cat (46)	chair	48	airplane (60)
fox	70	cat (43)	bowl	47	cat (23)
leopard	70	cat (41)	mushroom	46	bird (61)
camel	62	horse (31)	clock	45	ship (31)
tractor	54	truck (89)	bottle	40	cat (35)
motorcycle	36	automobile (44)	keyboard	30	airplane (33)

Table 21: Agreement between the four post-hoc scores on the Vanilla model: Spearman correlation of $u(x)$ and the share of unknowns (%) that only one score of the pair rejects.

Pair (A vs. B)	Spearman ρ of $u(x)$			Near: only A / only B rejects		Far: only A / only B rejects	
	All	Known test	Unknown	A	B	A	B
MSP vs. MLS	0.960	0.951	0.915	4.4	8.4	5.0	14.4
MSP vs. Energy	0.958	0.950	0.884	5.2	10.4	5.6	16.4
MSP vs. Mahalanobis	0.696	0.569	0.922	3.8	3.9	2.8	11.4
MLS vs. Energy	1.000	1.000	0.996	0.9	2.0	0.6	2.0
MLS vs. Mahalanobis	0.586	0.429	0.819	9.4	5.5	10.8	10.0
Energy vs. Mahalanobis	0.584	0.427	0.789	11.1	6.1	12.5	10.4